

Unit 6

SECURITY, PRIVACY AND DATA INTEGRITY

70 items · 365 marks

Syllabus sections in this unit

6.1 Data Security

6.2 Data Integrity

Unit 6: Security, privacy and data integrity

1. M/J 26/P11/Q4/d

- 4 A shop allows customers to pay to borrow toys. The shop uses a database to record information about the customers and the toys they have borrowed.

CUSTOMER (CustomerID, ContactFirstName, Phone)

TOY (ToyID, ToyName, CostPerDay)

TOY_INSTANCE (UniqueToyID, ToyID, Quality)

BORROW (BorrowID, CustomerID, UniqueToyID, StartDate, NumberDays)

- (d) When data is entered into the database it is validated and verified.

- (i) Explain how validation and verification support data integrity within a database.

[4]

- (ii) Give **one** reason why data may still be incorrect even after it passes validation and verification.

[1]

2. M/J 26/P12/Q2/a,b,c

- 2 Personal data is entered into a computer using a keyboard. The data is validated and verified during data entry.

- (a) Describe **two** methods of data verification that can be used during data entry.

1

2

[4]

- (b) The personal data includes the date of birth, for example 05/07/2010.

Complete the table by identifying **two** methods of validation that can be used on the date of birth.

State how each validation method will be used for the date of birth.

data validation method	use
.....	
.....	

[3]

(c) The data needs to be kept secure whilst on the computer.

(i) A virus is a threat to data.

Complete the table by describing what is meant by a virus.

Identify **and** describe **one** method to restrict the risk posed by a virus.

virus description	
method to restrict risk	
description of method to restrict risk	

[3]

(ii) A hacker is another threat to data.

Complete the table by describing what is meant by a hacker.

Identify **and** describe **one** method to restrict the risk posed by a hacker.

hacker description	
method to restrict risk	
description of method to restrict risk	

[3]

3. M/J 26/P13/Q1

- 1** Validation and verification can be used to protect the integrity of data.

Complete the description of methods of ensuring data integrity by writing the missing terms.

..... is a method of verification where the data is entered by a user and then manually compared to the original data to make sure that they are the same.

A verification method used during data transfer calculates a value from the data which is transferred with the data. The receiver performs the same calculation and compares the result with the value transferred.

This method of verification is

There are several methods of data validation.

check compares the data entered to a list of possible values that can be entered. If the data is not one of these values it is rejected.

..... check can be used to make sure data entered has the correct combination of letters, numbers and symbols. For example, a code may need to be two capital letters, one number followed by one symbol.

..... check is used to make sure the correct quantity of characters has been entered. For example, a name must be more than one character and less than or equal to 100 characters.

[5]

4. O/N 25/P11/Q8/a

- 8** One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(a) Viruses and pharming are examples of malware.

Explain what is meant by a virus and pharming.

Virus

Pharming

[2]

5. O/N 25/P12/Q1

- 1 Draw **one** line from each verification method to indicate whether it is used during data transfer or data entry.

Verification Method	Use
Parity byte check	
Checksum	Data transfer
Visual check	
Parity block check	Data entry

[2]

6. O/N 25/P12/Q5/e

- 5 A local area network (LAN) has five computers, one switch and one server.

- (e) Complete the table by identifying **one** threat to computer and data security posed by networks and the internet.

Describe the threat and give a method of prevention.

Threat	Description	Prevention

[3]

7. O/N 25/P12/Q6/c

- (c) A computer system uses even parity. The least significant (rightmost) bit of each byte is the parity bit.

- (i) Complete the byte by writing the missing parity bit:

							parity bit ↓
0	1	0	1	1	1	0	

[1]

- (ii) The computer also uses parity block check. The parity block check uses even parity. Computer A transmits four bytes of data to computer B, followed by a parity byte. Computer B receives the following sequence of bytes.

							parity bit ↓
1	0	1	1	0	1	1	1
0	1	1	1	0	0	0	0
0	0	0	1	1	0	1	1
0	1	1	1	0	1	0	0
parity byte →	1	0	1	0	0	0	0

Following transmission, one of the four bytes of data has an error in one of the bits.

Circle the bit that has been altered during the data transfer.

[1]

8. O/N 25/P12/Q10/c

- 10 A programmer is developing a computer program.

- (c) A user downloads the computer program from the internet.

State what should be included as part of the download to make sure the program is authentic.

[1]

9. O/N 25/P13/Q7/c,e

- 7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

- (c) Identify **and** describe **one** method of data verification that can be used when transferring data from the digital video camera to the server.

Method

Description

[3]

- (e) The table contains three algorithms that perform data validation. The third algorithm uses the functions `MID` and `LENGTH`.

The functions `MID` and `LENGTH` are defined as follows:

- `MID(ThisString : STRING, x : INTEGER, y : INTEGER)` RETURNS STRING
returns a string of length `y` starting at position `x` from `ThisString`
- `LENGTH(ThisString : STRING)` RETURNS INTEGER
returns the integer value representing the length of `ThisString`

Complete the table to identify the method of data validation for each algorithm. Each method of data validation must be different.

Algorithm	Method of data validation
<pre> INPUT x IF x < 1 or x > 26 THEN OUTPUT "Invalid" ENDIF </pre>	
<pre> INPUT x IF x <> 'R' AND x <> 'G' AND x <> 'B' THEN OUTPUT "Invalid" ENDIF </pre>	
<pre> INPUT x FLAG ← FALSE FOR INDEX ← 1 TO LENGTH(x) IF MID(x, INDEX, 1) = "@" THEN FLAG ← TRUE ENDIF NEXT INDEX IF NOT FLAG THEN OUTPUT "Invalid" ENDIF </pre>	

10. O/N 25/P21/Q6/a

- 6 Students are learning about a simple check digit method for data validation. In this method, a single check digit is appended to the end of an original number to give a new number.

The students are studying a method which:

- calculates the sum of all the digits in the original number
- uses integer division to calculate the remainder when the sum is divided by 10
- uses the remainder as the check digit
- appends the check digit to the original number, creating the new number.

For example:

original number	4162
sum of all digits	$4 + 1 + 6 + 2 = 13$
remainder when the sum is divided by 10 using integer division	3
new number	41623

The method described can be used to detect single-digit errors. For example, if the new number is incorrectly input as 41633, then the check digit does not match and the input will be rejected.

- (a) An incorrect attempt was made to enter 41623. The **first two** digits were entered incorrectly; the **last three** digits were entered correctly.

When this number was tested, the check digit was found to be correct and the input was accepted.

Identify an example of the incorrect attempt to enter 41623 and explain why this number would **not** be rejected.

Number

Explanation

[2]

11. M/J 25/P11/Q7/a,b,c

7 A banker stores personal data on their work computer.

(a) The banker needs to transfer confidential data across the internet.

Identify and describe **one** method of restricting the risks posed by an unauthorised person intercepting the data whilst it is being transferred across the internet.

Method

Description

[3]

(b) The banker receives confidential data across the internet. The data includes a digital signature.

Explain how a digital signature can make sure the data has **not** been changed during transmission.

[5]

(c) The data that is transferred can also be verified using a checksum.

Explain how data can be verified using a checksum.

[3]

12. M/J 25/P13/Q3/b

3 A programmer is writing a program in a high-level language.

- (b)** The programmer needs to keep the program files secure on their computer and during electronic data transmission over the internet. The files are protected by a password.

Complete the table by identifying **one** other method of keeping the files secure during electronic data transmission **and one** other method of keeping the files secure on the computer.

State how each method protects the data.

The methods must be different.

	Method	How the method protects the data
during transmission		
on the computer		

[4]

13. O/N 24/P12/Q4/c

4 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

- (c)** The file containing the final program code will be sent by email for beta testing.

Identify **one** security method that can be used to protect the program code from unauthorised access during email transfer.

Explain how your chosen method protects the program code.

Security method

Explanation

[3]

14. M/J 24/P11/Q5/b,c(i)

- 5** A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

- (b)** The bank wants to protect the integrity of its data while transferring the data to other banks. Parity check is one example of data verification.

Complete the description of parity check when Computer A is transmitting data to Computer B.

Computer A and Computer B agree on whether to use

parity. Computer A divides the data into groups of The

number of 1s in each group is counted. If the agreed parity is

and the group has an even number of 1s, a parity bit of 1 is appended, otherwise a parity bit of 0 is appended.

In a parity check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity

[5]

- (c)** The bank also needs to keep its customers' data private and secure.

- (i)** The bank's network has a firewall.

Explain how a firewall can help protect the customers' data.

[3]

15. M/J 24/P12/Q3/a

- 3** An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(a) The assessment board needs to make sure the data stored on the server is secure.

- (i)** Authentication methods can help to protect the server against hackers.

Identify **one other** security measure that helps to protect the server from hackers.

Describe how the security measure works.

Security measure

Description

[3]

- (ii)** Identify **one** security measure that helps to protect the data when it is being transmitted to its destination. Describe how the security measure works.

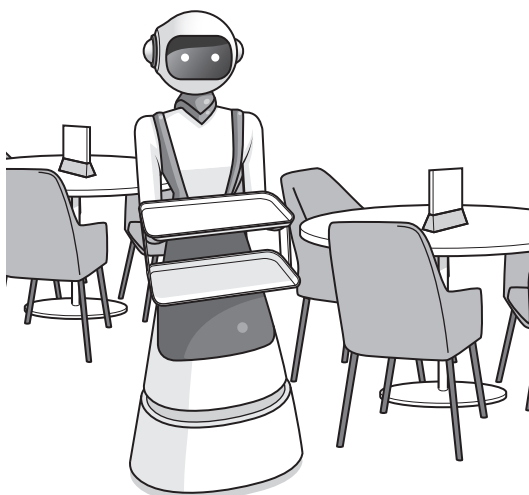
Security measure

Description

[3]

16. M/J 24/P13/Q7/f

- 7** Robots are used to serve food and drink to customers at a restaurant.



(f) The data from the robots is transmitted to a central computer using a wireless connection.

- (i)** Complete the table by identifying **and** describing **two** methods of data verification that can be used during data transfer.

	Method	Description
1		
2		

[4]

(ii) Explain how encryption can protect the security of data during transmission.

[2]

17. O/N 23/P12/Q5/a,b,c

5 (a) State the meaning of **privacy of data**.

[1]

(b) State the meaning of **integrity of data**.

[1]

(c) Describe the following threats to a computer system.

Phishing email

Spyware

[4]

18. O/N 23/P13/Q3/a(iii),c

- 3** A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (iii)** State how the following security measures can be used to protect computer systems.

Firewall

Encryption

Passwords

[3]

- (b)** The company uses a database, `COURSES`, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

The database has the following structure:

`COURSE_INFORMATION(CourseID, Description, Cost)`

`TUTOR(TutorID, TelephoneNumber, EmailAddress, TutorName)`

`COURSE_SCHEDULE(CourseID, DateStarted, TutorID)`

- (c)** An example of a tutor ID is NK16C6.

An administrative officer enters the tutor ID into the `TUTOR` table.

Explain how data verification can be used when the tutor ID is entered.

[4]

19. O/N 23/P13/Q8/a,b,c

- 8** **(a)** Data verification is one method of protecting the integrity of data.

Describe **one** other method of protecting the integrity of data.

[2]

- (b)** State **one** difference and **one** similarity between pharming and phishing.

Difference

Similarity

[2]

- (c)** Explain how the data security risks of malware can be restricted.

[3]

20. M/J 23/P12/Q2/c(i)

- 2** A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

- (c)** The database `Lessons` has the following tables:

`HORSE(HorseID, Name, Height, Age, HorseLevel)`

`STUDENT(StudentID, FirstName, LastName, RiderLevel, PreferredHorseID)`

`LESSON(LessonID, Date, Time, StudentID, HorseID, LessonContent)`

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

- (i)** Describe **two** methods of validating the field `RiderLevel`.

1

2

[2]

21. M/J 23/P13/Q4/d

- 4** A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(d) The data is validated and verified when it is entered into the database.

- (i)** The car registration number must be: 1 letter, followed by 3 numbers, followed by 2 letters.

For example, A123AA is valid but A12AA is invalid.

One way that a registration number can be validated is by using a presence check to make sure the registration number has been entered.

Describe **two other** ways that the car registration number can be validated.

1

2

[2]

- (ii)** Describe **two** ways that the car registration number can be verified when it is entered into the database.

1

2

[2]

- (iii)** State why the car registration number might be incorrect even after it has been validated and verified.

[1]

22. M/J 23/P13/Q6/a,b

6 Data needs to be kept secure when stored on a computer and during transmission over a network.

- (a) Explain how a digital signature is used to authenticate a digital document during transmission over a network.

[5]

- (b) Complete the table by identifying **and** describing **two** types of software that can be installed on a computer to prevent threats over a network.

Type of software	Description
.....	
.....	

[2]

23. M/J 23/P23/Q2/b

2 A program stores a date of birth for a student using a variable, `MyDOB`, of type `DATE`.

- (b) A function `CheckDate()` will take three integer parameters representing a day, month and year of a given date.

The function will validate the date of birth for a student that the parameters passed to it represent.

For a date to be valid, a student must be at least 18 in year 2020.

- (i) Two of the parameter values can be checked without reference to the third parameter.

Describe these **two** checks.

Check 1

Check 2

[2]

- (ii) Several values of the parameter representing the day can only be checked completely by referring to the value of **one other** parameter.

Describe this check.

[2]

24. O/N 22/P11/Q2

2 Draw **one** line from each security feature to its most appropriate description.

Security feature	Description
firewall	converts data to an alternative form
pharming	redirects a user to a fake website
anti-virus software	verifies the authenticity of data
encryption	scans files on the hard drive for malicious software
	accepts or rejects incoming and outgoing packets based on criteria

[4]

25. O/N 22/P12/Q4/a,b,c

4 (a) State the difference between **data verification** and **data validation**.

[1]

(b) A checksum can be used to detect errors during data transmission.

Describe how a checksum is used.

[3]

(c) One validation method is a presence check.

Describe **two other** validation methods that can be used to validate non-numeric data.

1

2

[2]

26. O/N 22/P12/Q6/a

6 (a) A student uses a networked laptop computer to send an email to a colleague.

(i) Explain how a digital signature ensures the email is authentic.

[2]

(ii) Describe how a firewall protects the data on the computer.

[3]

27. O/N 21/P11/Q2/a,b,c

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(a) Explain the difference between data security and data integrity.

[2]

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(c) Two malware threats are spyware and viruses.

Give **two** similarities and **one** difference between spyware and a virus.

Similarity 1

Similarity 2

Difference

[3]

28. O/N 21/P12/Q1

- 1 When designing computer systems, it is important to consider the security, integrity and privacy of the data.

Draw **one** line from each measure to indicate whether it keeps data secure or protects the integrity of data.

Measure

Firewall

Double entry

Presence check

Access rights

Password

Data Security

Data Integrity

[2]

29. O/N 21/P13/Q2/a,b,c

2 Xanthe wants to maintain the integrity and security of data stored on her computer.

(a) Explain the difference between data security and data integrity.

[2]

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description

Example

[2]

(c) Two malware threats are spyware and viruses.

Give **two** similarities and **one** difference between spyware and a virus.

Similarity 1

Similarity 2

Difference

[3]

30. M/J 21/P11/Q6/a,b,c

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(a)

```
INPUT x
IF x < 0 OR x > 10 THEN
    OUTPUT "Invalid"
ENDIF
```

[1]

(b)

```
INPUT x
IF x = "" THEN
    OUTPUT "Invalid"
ENDIF
```

[1]

(c)

```
INPUT x
IF NOT(x = "Red" OR x = "Yellow" OR x = "Blue") THEN
    OUTPUT "Invalid"
ENDIF
```

[1]

31. M/J 21/P12/Q8/a,b,c

- 8** A company has several security measures in place to prevent unauthorised access to the data on its computers.

(a) Describe the difference between the security and privacy of data.

[2]

- (b)** Each employee has a username and password to allow them to log onto a computer. An employee's access rights to the data on the computers is set to either read-only, or read and write.

Identify **one** other software-based measure that could be used to restrict the access to the data on the computers.

[1]

- (c)** The company is also concerned about threats posed by networks and the internet.

Identify **two** threats to the data that are posed by networks and the internet.

Threat 1

Threat 2

[2]

32. M/J 21/P13/Q6/a,b,c

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(a)

```
INPUT x  
  
IF x < 0 OR x > 10 THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(b)

```
INPUT x  
  
IF x = "" THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

(c)

```
INPUT x  
  
IF NOT(x = "Red" OR x = "Yellow" OR x = "Blue") THEN  
  
    OUTPUT "Invalid"  
  
ENDIF
```

[1]

33. 9608 M/J 21/P11/Q2/d

- 2** Zak designs a logo for his company. He uses vector graphics software to create the logo.



- (d)** Zak's company holds details about clients in a database.

Give **three** security measures that Zak can implement to make sure that only authorised employees can access the data.

- 1
- 2
- 3

[3]

34. 9608 M/J 21/P12/Q5/e(ii)

5 A web page includes HTML and JavaScript code.

```
01 <html>
02 <body>
03
04 Calculate area of a triangle:
05 <form name = "Triangle">
06   <p>Base <input type = "number" name = "B" value = ""></p>
07   <p>Height <input type = "number" name = "H" value = ""></p>
08   <button onclick = "area()">Calculate</button>
09 </form>
10
11 <script>
12 function area() {
13   var base = document.forms["Triangle"]["B"].value;
14   var height = document.forms["Triangle"]["H"].value;
15   if (base == "" || height == "") {
16     alert("Both values must be entered");
17     return false;
18   }
19   else {
20     area = 0.5 * height * base;
21     alert("The area is: " + area);
22   }
23 }
24 </script>
25 </body>
26 </html>
```

(e) Data validation has been used in line 15 of the JavaScript code.

(ii) Identify **two** other data validation checks that could be used.

1

2

[2]

Cambridge International Holidays allows customers to make holiday bookings on its website.

35. 9608 M/J 21/P12/Q8/a,b

- 8 (a) Complete the following table by identifying the **most appropriate** term for each description. Each term must be different.

Description	Term
Ensures data is accurate and up to date	
Prevents accidental or malicious data loss	
Prevents unauthorised access to data	

[3]

- (b) Describe what is meant by a **digital signature**.

[2]

36. 9608 M/J 21/P13/Q1/a,b

1 An adventure sports company has a website.

- (a)** Customers can book courses using an online booking form. The booking form contains a number of fields.

The following table gives a description of the validation for each field.

Write the validation type for each validation description in the table.

Field	Validation description	Validation type
Name	A name must be entered	
Date of Birth	Entered as dd/mm/yyyy	
Telephone Number	A limit of 15 characters can be entered	
Experience Level	Only values between 1 and 5 can be entered	

[4]

- (b) (i)** Validation is one way to protect the integrity of input data.

Identify **one other** method to protect the integrity of input data.

[1]

- (ii)** The data input will be transferred to a central server.

Identify **two** measures to protect the integrity of the data during transfer.

1

2

[2]

37. 9608 M/J 21/P21/Q5/a(ii)

- 5** Alice is developing an algorithm for a game. The game will count the number of attempts by a player to guess the secret number generated by the built-in random number function.

She will use pseudocode to develop a procedure, `GuessNum()`, that will:

1. generate a secret number that is a random integer in the range 1 to 20 inclusive
2. prompt and input the player's guess
3. output a message if the guess is not the same as the secret number and repeat from step 2
4. count the number of attempts it takes to guess the secret number
5. output a message when the secret number has been guessed correctly, giving the number of attempts.

- (ii)** Suggest **two** different validation checks that could be applied to the input of the procedure `GuessNum()` to ensure invalid guesses are not counted.

1

2

[2]

38. M/J 20/P11/Q1/a,b

1 A computer program makes use of data validation routines and verification of data input.

(a) Complete these **two** sentences about data validation and verification.

1. checks that the data entered is reasonable. One example is

2. checks that the data entered is the same as the original. One example is

[4]

(b) The program is installed on a computer system that has security measures in place to protect its data.

Complete the following table.

Security measure	Description
.....	Data are written on two or more disks simultaneously.
Encryption	
.....	A copy of the data is taken and stored in another location.

[3]

39. M/J 20/P12/Q7/d

7 A company allows customers to stream music from its servers over the Internet. The company's internet connection is currently provided through copper cables.

(d) The company needs to keep the data on its servers secure from online threats.

(i) Describe how a firewall will help to protect the data on the servers from online threats.

[2]

(ii) Give **one additional** security measure that the company can use to protect the data on the servers from online threats.

[1]

40. M/J 20/P13/Q5/a

5 Wei is developing a program.

(a) He wants to make sure the source code is secure on his laptop.

Explain how encrypting the source code can keep it secure.

[3]

41. O/N 19/P13/Q1/c(ii)

1 In a supermarket, a self-checkout machine allows customers to scan the barcodes of products and then pay for their shopping. These are an alternative to the traditional cashier-staffed checkout.



(c) The self-checkout machines connect to a server that stores all the data for the supermarket. This is a client-server network.

(ii) The supermarket is concerned about the security and integrity of the data on the server.

Identify **two** methods that can be used to minimise the security risk to the data, and **one** method to protect the integrity of the data.

Security 1

Security 2

Integrity

[3]

42. M/J 19/P12/Q5/b(i)

5 Moheem is creating a relational database to store data about his customers.

(b) Moheem uses a Database Management System (DBMS) to ensure the security and integrity of the data.

(i) Explain the difference between security and integrity.

[2]

43. M/J 19/P13/Q2/a,b,c

2 Frankie is a software developer. He is developing a program to manage customer records for a client with an online retail business. He must ensure that data stored about each customer are both secure and private.

(a) State the difference between security and privacy.

[2]

(b) Computer systems can be protected by physical methods such as locks.

Describe **two** non-physical methods used to improve the security of computer systems.

1

2

[6]

(c) A computer uses parity blocks to check the data that has been received is the same as the data that has been transmitted.

The following is an example of a parity block.

	Parity bit	Data						
	1	1	1	1	0	0	0	1
	0	0	0	0	1	1	1	0
	1	1	0	1	1	0	0	1
Parity byte	1	1	0	1	1	0	0	1

(i) Describe how a parity block check can identify a bit that has been corrupted during transmission.

[4]

(ii) Give a situation where a parity block check **cannot** identify corrupted bits.

[1]

44. O/N 18/P12/Q6/a

6 Kim is using her laptop computer to write a program in a high-level language.

- (a)** Kim needs to make sure the program is secure against unauthorised access. She has already set up a username and password on her laptop.

Identify **two** additional electronic measures that Kim can use to keep the program secure.

1

2

[2]

45. O/N 18/P13/Q5/b

5 Arnold is a software developer. He has created a computer game for people to download over the Internet. Arnold is considering releasing the game as a piece of commercial software.

- (b)** Users need to enter their name and email address to create an account. The information is stored in a database on Arnold's computer.

Give **three** ways that Arnold can ensure users' details are kept secure.

1

2

3

[3]

46. M/J 18/P11/Q6/a,b,c,d

6 Parity bits can be used to verify data.

(a) The following binary number is transmitted using **even** parity.

Add the missing parity bit.

**Parity
bit**

	1	0	1	1	0	1	0
--	---	---	---	---	---	---	---

[1]

(b) In the following parity block, the first column contains the parity bits, and the last row contains the parity byte. A device transmits the data using **even** parity.

(i) Circle the error in the data transmitted.

	Parity bit	Data						
	1	1	0	1	0	1	1	1
	1	0	0	0	1	1	1	0
	0	1	0	0	1	0	1	1
	1	1	1	0	1	1	1	1
Parity byte	1	1	1	1	1	0	0	1

[1]

(ii) Explain how you identified the error.

[2]

(c) The data received can contain errors that are not detected using parity bits.

Explain how this can happen.

[2]

(d) Parity is not the only method to verify the data has been sent correctly.

Name **and** describe **one** other method of data verification during data transfer.

Name

Description

[3]

47. M/J 18/P12/Q3/a,b,c

3 Parity bits can be used to verify data.

(a) The following binary number is transmitted using **odd** parity.

Add the missing parity bit.

**Parity
bit**

	0	1	0	0	0	0	0
--	---	---	---	---	---	---	---

[1]

(b) In the following data transmitted, the first column contains the parity bits, and the last row contains the parity byte. A device transmits the data using **even** parity.

Circle the error in the data transmitted.

	Parity bit	Data						
	1	0	1	0	1	1	1	1
	0	1	1	0	0	1	1	0
	1	1	0	0	0	0	0	0
	0	1	0	0	0	0	0	0
Parity byte	0	0	0	0	1	0	0	1

[1]

(c) The following table shows five error detection measures.

Put **one** tick (✓) in each row to indicate whether the measure is validation or verification.

Measure	Validation	Verification
Checksum		
Format check		
Range check		
Double entry		
Check digit		

[5]

48. M/J 18/P13/Q4/a,b,c

4 Parity bits can be used to verify data.

(a) The following binary number is transmitted using **odd** parity.

Add the missing parity bit.

**Parity
bit**

	0	1	1	1	0	1	0
--	---	---	---	---	---	---	---

[1]

(b) In the following parity block, the first column contains the parity bits, and the last row contains the parity byte. A device transmits the data using **even** parity.

Circle the error in the data being transmitted.

	Parity bit	Data						
	0	0	1	1	0	1	0	1
	1	1	1	1	1	0	0	1
	1	0	1	0	1	0	0	0
	0	0	0	1	1	0	1	1
Parity byte	0	1	1	1	1	1	0	1

[1]

(c) Four error detection measures are shown.

Draw **one** line from each error detection measure to indicate whether it is verification or validation.

Error detection measure

Type check

Proof reading

Check digit

Checksum

Verification

Validation

[4]

49. O/N 17/P12/Q3/a,b,c

- 3** A Local Area Network is used by school staff who access data stored in a Database Management System (DBMS).

(a) (i) Explain the difference between security and privacy of data.

[3]

(ii) Give an example for this application where privacy of data is a key concern.

[1]

- (b)** Name and describe **two** security measures the Network Manager has in place to protect the security of the data held in the DBMS.

1

2

[4]

- (c)** A task for staff at the start of the school year is to key in new pupil data from a paper document.

The data is entered to a screen form and includes the data verification of some fields.

Describe what is meant by **verification**.

[2]

50. O/N 17/P13/Q3/a,b

- 3** A Local Area Network is used by staff in a hospital to access data stored in a Database Management System (DBMS).

(a) Name **two** security measures to protect computer systems.

1

2

[2]

- (b)** A frequent task for staff is to key in new patient data from a paper document. The document includes the patient's personal ID number.

- (i)** The Patient ID is a seven digit number. The database designer decides to use a check digit to verify each foreign key value that a user keys in for a Patient ID.

When a user assigns a primary key value to a Patient ID, the DBMS adds a modulus-11 check digit as an eighth digit. The DBMS uses the weightings 6, 5, 4, 3, 2 and 1 for calculating the check digit. It uses 6 as the multiplier for the most significant (leftmost) digit.

Show the calculation of the check digit for the Patient ID with the first six digits 786531.

Complete Patient ID

[4]

- (ii)** Name and describe **two** validation checks that the DBMS could carry out on each primary key value that a user keys in for a Patient ID.

1 Validation check

Description

2 Validation check

Description

[4]

51. M/J 17/P11/Q5/a,b

- 5** A computer receives data from a remote data logger. Each data block is a group of 8 bytes. A block is made up of seven data bytes and a parity byte.

Each data byte has a parity bit using odd parity. The parity byte also uses odd parity.

The following table shows a data block before transmission. Bit position 0 is the parity bit.

Bit position							
7	6	5	4	3	2	1	0
1	1	0	0	1	1	0	1
0	0	1	0	0	0	0	0
1	0	0	1	1	1	0	A
1	1	0	0	0	0	1	0
1	1	0	0	0	0	1	0
1	1	0	0	0	1	1	B
0	0	0	0	0	0	0	0

Data bytes

0	1	1	0	1	1	0	0
---	---	---	---	---	---	---	---

Parity byte

- (a) (i) Describe how the data logger calculates the parity bit for each of the bytes in the data block.

[2]

- (ii) State the two missing parity bits labelled **A** and **B**.

A =

B =

[1]

- (iii) Describe how the computer uses the parity byte to perform a further check on the received data bytes.

[2]

- (b) (i) A second data block is received as shown in the following table. There are errors in this data block.

Identify and then circle **two** bits in the table which must be changed to remove the errors.

Bit position							
7	6	5	4	3	2	1	0
1	0	0	0	1	1	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	1	0	1
1	1	1	1	0	0	0	1
1	1	0	0	0	0	1	0
0	0	1	0	0	1	0	0
0	0	0	0	0	0	0	1

0	1	0	1	1	0	0	0
---	---	---	---	---	---	---	---

[2]

- (ii) Explain how you arrived at your answers for **part (b)(i)**.

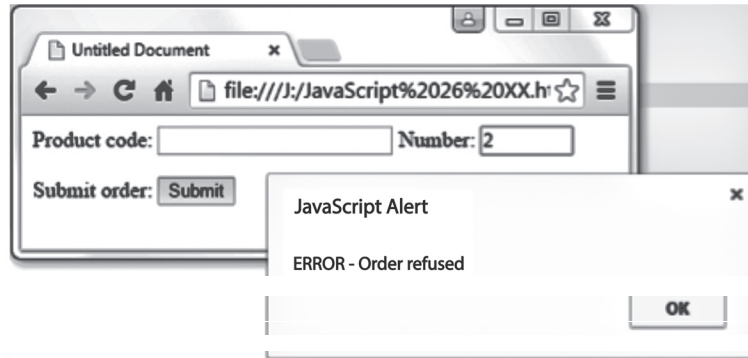
[3]

52. M/J 17/P11/Q7/c(v)

7 The design of a web-based application can require the use of client-side scripting.

(c) A web page used for data capture consists of:

- two text boxes for the entry of:
 - a product code
 - the number of items to be purchased.
- a button which is clicked when the user wants to submit this order.



Study the following web page.

```

1 <html>
2 <head>
3 <title>Untitled Document</title>
4 <script language="JavaScript">
5
6 function myButton_onmousedown()
7 {
8   var Message1 = "ERROR - Order refused";
9   var Message2 = "Product code OK";
10  var x = document.forms["form1"]["txtProductCode"].value;
11    if (x == "")
12    {
13      alert(Message1)
14    }
15    else
16    {
17      alert(Message2)
18    }
19  }
20 </script>
21
22 </head>
23 <body>
24 <form name = form1>
25   <label>Product code: </label>
26   <input type="text" name="txtProductCode" >
27   <label>Number: </label>
28   <input type="text" name="txtNumber" size = "5" >
29   <p>
30     <label>Submit order: </label>
31     <input type="button" name="btnSubmit" Value = "Submit"
32
33     onMouseDown = "myButton_onmousedown()" >
34   </p>
35 </form>
36
37 </body>
38 </html>

```

- (v) Name and describe **two** other types of validation check that could be appropriate for this data capture form.

Validation check:

Description

Validation check:

Description

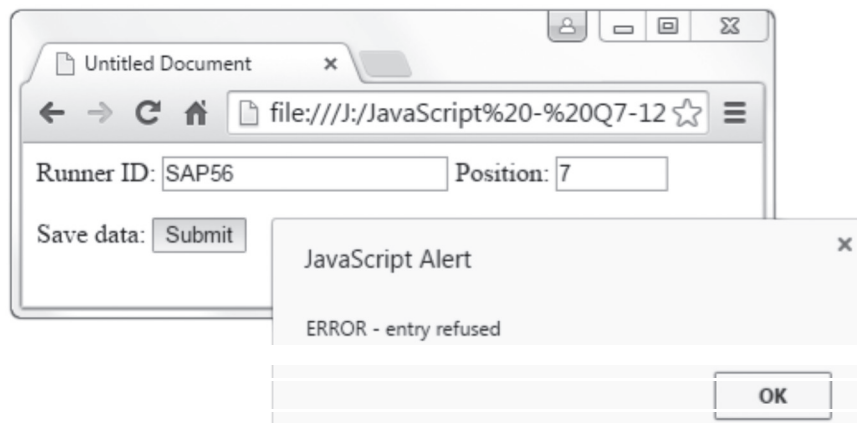
[4]

53. M/J 17/P12/Q6/c(v)

- 6 Downloading a file from a website is an example of a client-server application.

(c) The following web page used for data capture consists of:

- two text boxes for the entry of:
 - a race runner's ID code
 - their finishing position in a race.
- a button that the user clicks to submit this runner's result.



```

1  <html>
2  <head>
3  <title>Untitled Document</title>
4  <script language="JavaScript">
5
6  function myButton_onmousedown()
7  {
8  var Output1 = "Runner ID OK";
9  var Output2 = "ERROR - entry refused";
10
11 var Runner ID = document.forms["form1"]["txtRunnerID"].value;
12 //                                     || in Javascript is the 'OR' operator
13 if (RunnerID.substr(0,3) == "VAR" || RunnerID.substr(0,3) == "CAM")
14 {
15     alert(Output1)
16 }
17 else
18 {
19     alert(Output2)
20 }
21 }
22 </script>
23
24 </head>
25 <body>
26 <form name = form1>
27   <label>Runner ID: </label>
28   <input type="text" name="txtRunnerID" >
29   <label>Position: </label>
30   <input type="text" name="txtPosition" size = "5" >
31   <p>
32     <label>Save data: </label>
33     <input type="button" name="btnSubmit" Value = "Submit"
34
35     onMouseDown = "myButton_onmousedown()" >
36   </p>
37 </form>
38
39 </body>
40 </html>

```

- (v) Name and describe **two** other types of validation check which could be appropriate for this data capture form.

Validation check:

Description

Validation check:

Description

[4]

54. M/J 17/P13/Q5/a,b

- 5 A computer receives data from a remote data logger. Each data block is a group of 8 bytes. A block is made up of seven data bytes and a parity byte.

Each data byte has a parity bit using odd parity. The parity byte also uses odd parity.

The following table shows a data block before transmission. Bit position 0 is the parity bit.

Bit position							
7	6	5	4	3	2	1	0
1	1	0	0	1	1	0	1
0	0	1	0	0	0	0	0
1	0	0	1	1	1	0	A
1	1	0	0	0	0	1	0
1	1	0	0	0	0	1	0
1	1	0	0	0	1	1	B
0	0	0	0	0	0	0	0

Data bytes

0	1	1	0	1	1	0	0
---	---	---	---	---	---	---	---

← Parity byte

- (a) (i) Describe how the data logger calculates the parity bit for each of the bytes in the data block.

[2]

- (ii) State the two missing parity bits labelled **A** and **B**.

A =

B =

[1]

- (iii) Describe how the computer uses the parity byte to perform a further check on the received data bytes.

[2]

- (b) (i) A second data block is received as shown in the following table. There are errors in this data block.

Identify and then circle **two** bits in the table which must be changed to remove the errors.

Bit position							
7	6	5	4	3	2	1	0
1	0	0	0	1	1	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	1	0	1
1	1	1	1	0	0	0	1
1	1	0	0	0	0	1	0
0	0	1	0	0	1	0	0
0	0	0	0	0	0	0	1

0	1	0	1	1	0	0	0
---	---	---	---	---	---	---	---

[2]

(ii) Explain how you arrived at your answers for **part (b)(i)**.

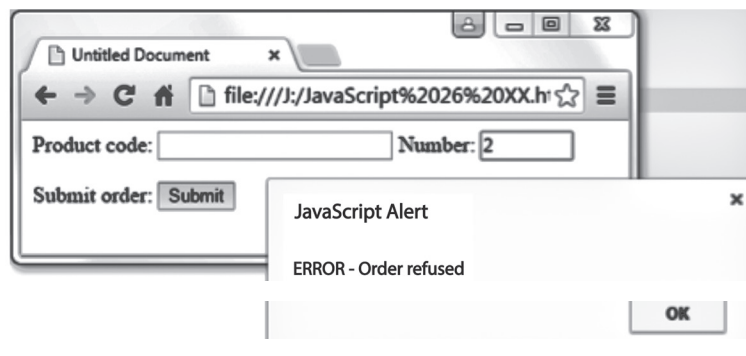
[3]

55. M/J 17/P13/Q7/c(v)

7 The design of a web-based application can require the use of client-side scripting.

(c) A web page used for data capture consists of:

- two text boxes for the entry of:
 - a product code
 - the number of items to be purchased.
- a button which is clicked when the user wants to submit this order.



Study the following web page.

```
1 <html>
2 <head>
3 <title>Untitled Document</title>
4 <script language="JavaScript">
5
6 function myButton_onmousedown()
7 {
8   var Message1 = "ERROR - Order refused";
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10  var x = document.forms["form1"]["txtProductCode"].value;
11    if (x == "")
12    {
13      alert(Message1)
14    }
15    else
16    {
17      alert(Message2)
18    }
19  }
20 </script>
21
22 </head>
23 <body>
24 <form name = form1>
25   <label>Product code: </label>
26   <input type="text" name="txtProductCode" >
27   <label>Number: </label>
28   <input type="text" name="txtNumber" size = "5" >
29   <p>
30     <label>Submit order: </label>
31     <input type="button" name="btnSubmit" Value = "Submit"
32
33     onMouseDown = "myButton_onmousedown()" >
34   </p>
35 </form>
36
37 </body>
38 </html>
```

- (v) Name and describe **two** other types of validation check that could be appropriate for this data capture form.

Validation check:

Description

Validation check:

Description

56. O/N 16/P11/Q7/c

7 A small company produces scientific magazines. The owner buys some new desktop computers. The computers are used to store thousands of colour images (diagrams and photographs). All the computers have Internet access.

- (c)** Employees using the new computers receive training. At the end of the training, each employee completes a series of questions.

Three answers given by an employee are shown below.

Explain why each answer is incorrect.

- (i)** *“Encryption prevents hackers breaking into the company’s computers.”*

[2]

- (ii)** *“Data validation is used to make sure that data keyed in are the same as the original data supplied.”*

[2]

- (iii)** *“The use of passwords will always prevent unauthorised access to the data stored on the computers.”*

[2]

57. O/N 16/P13/Q7/c

7 A small company produces scientific magazines. The owner buys some new desktop computers. The computers are used to store thousands of colour images (diagrams and photographs). All the computers have Internet access.

- (c)** Employees using the new computers receive training. At the end of the training, each employee completes a series of questions.

Three answers given by an employee are shown below.

Explain why each answer is incorrect.

- (i)** *“Encryption prevents hackers breaking into the company’s computers.”*

[2]

- (ii)** *“Data validation is used to make sure that data keyed in are the same as the original data supplied.”*

[2]

- (iii)** *“The use of passwords will always prevent unauthorised access to the data stored on the computers.”*

[2]

58. M/J 16/P13/Q7/a,b

7 A bank holds personal data about its customers and their financial data.

(a) Describe the difference between security and integrity of data.

[4]

(b) Describe **three** security measures that the bank could implement to protect its electronic data.

Security measure 1

Description

Security measure 2

Description

Security measure 3

Description

[6]

59. O/N 15/P11/Q9/a,b

9 (a) Give a brief description of each of the following terms:

Validation

Verification

[2]

(b) Data are to be transferred between two devices. Parity checks are carried out on the data.

Explain what is meant by a parity check. Give an example to illustrate your answer.

[4]

60. O/N 15/P11/Q10/a,b

10 (a) Explain the term computer virus.

[2]

(b) A virus checker has been installed on a PC.

Give **two** examples of when a virus checker should perform a check.

1

2

[2]

61. O/N 15/P13/Q9/a,b

9 (a) Give a brief description of each of the following terms:

Validation

Verification

[2]

(b) Data are to be transferred between two devices. Parity checks are carried out on the data.

Explain what is meant by a parity check. Give an example to illustrate your answer.

[4]

62. O/N 15/P13/Q10/a,b

10 (a) Explain the term computer virus.

[2]

(b) A virus checker has been installed on a PC.

Give **two** examples of when a virus checker should perform a check.

1

2

[2]

63. O/N 15/P21/Q7/c

7 ASCII character codes are used to represent a single character.

Part of the code table is shown below.

ASCII code table (part)

Character	Decimal	Character	Decimal	Character	Decimal
<Space>	32	I	73	R	82
A	65	J	74	S	83
B	66	K	75	T	84
C	67	L	76	U	85
D	68	M	77	V	86
E	69	N	78	W	87
F	70	O	79	X	88
G	71	P	80	Y	89
H	72	Q	81	Z	90

Some pseudocode statements follow which use the built-in functions below:

ONECHAR(ThisString : STRING, Position : INTEGER) RETURNS CHAR
 returns the single character at position Position (counting from the start of the string with value 1) from the string ThisString.
 For example: ONECHAR("Barcelona", 3) returns 'r'.

`CHARACTERCOUNT(ThisString : STRING)` RETURNS INTEGER
returns the number of characters in the string `ThisString`.
For example: `CHARACTERCOUNT("BRAZIL")` returns 6.

`CHR(ThisInteger : INTEGER)` RETURNS CHAR
returns the character with ASCII code `ThisInteger`.
For example: `CHR(65)` returns character 'A'.

`ASC(ThisCharacter : CHAR)` RETURNS INTEGER
returns the ASCII value for character `ThisCharacter`.
For example: `ASC('A')` returns 65.

- (b) A program is to be written which accepts a string and then calculates a numeric value from this string. The input string and the calculated value are then to be sent to a remote computer over a communications link.

Study the following pseudocode:

```
OUTPUT "Enter string"
INPUT MyString
StringTotal ← 0

FOR i ← 1 TO CHARACTERCOUNT(MyString)
    NextNum ← ASC(ONECHAR(MyString, i))
    StringTotal ← StringTotal + NextNum
ENDFOR

OUTPUT MyString, StringTotal
```

Write the above pseudocode algorithm as **program code**.

There is no need to show the declaration of variables or comment statements.

[6]

- (c) Explain the purpose of sending the value of `StringTotal` to the remote computer, in addition to `MyString`.

[2]

64. O/N 15/P23/Q7/c

- 7 ASCII character codes are used to represent a single character.

Part of the code table is shown below.

ASCII code table (part)

Character	Decimal	Character	Decimal	Character	Decimal
<Space>	32	I	73	R	82
A	65	J	74	S	83
B	66	K	75	T	84
C	67	L	76	U	85
D	68	M	77	V	86
E	69	N	78	W	87
F	70	O	79	X	88
G	71	P	80	Y	89
H	72	Q	81	Z	90

Some pseudocode statements follow which use the built-in functions below:

ONECHAR(ThisString : STRING, Position : INTEGER) RETURNS CHAR
 returns the single character at position Position (counting from the start of the string with value 1) from the string ThisString.
 For example: ONECHAR("Barcelona", 3) returns 'r'.

CHARACTERCOUNT(ThisString : STRING) RETURNS INTEGER
 returns the number of characters in the string ThisString.
 For example: CHARACTERCOUNT("BRAZIL") returns 6.

CHR(ThisInteger : INTEGER) RETURNS CHAR
 returns the character with ASCII code ThisInteger.
 For example: CHR(65) returns character 'A'.

ASC(ThisCharacter : CHAR) RETURNS INTEGER
 returns the ASCII value for character ThisCharacter.
 For example: ASC('A') returns 65.

- (b) A program is to be written which accepts a string and then calculates a numeric value from this string. The input string and the calculated value are then to be sent to a remote computer over a communications link.

Study the following pseudocode:

```

OUTPUT "Enter string"
INPUT MyString
StringTotal ← 0

FOR i ← 1 TO CHARACTERCOUNT(MyString)
    NextNum ← ASC(ONECHAR(MyString, i))
    StringTotal ← StringTotal + NextNum
ENDFOR

OUTPUT MyString, StringTotal
  
```

Write the above pseudocode algorithm as **program code**.

There is no need to show the declaration of variables or comment statements.

[6]

- (c) Explain the purpose of sending the value of `StringTotal` to the remote computer, in addition to `MyString`.

[2]

65. M/J 15/P11/Q4/b

- (b) Sensors are used to monitor seismic activity. At the end of each day, all the data are transmitted to a central computer. This is hundreds of kilometres away.

Describe **one** way of ensuring that the integrity of the data is retained during the transmission stage.

[4]

66. M/J 15/P11/Q6/b

- 6 (a) Name the **most** suitable input or output device for each of the following uses.

Give a different device in **each** case.

Description of use	Input or output device
input of credit card number into an online form	
selection of an option at an airport information kiosk	
output of a single high-quality photograph	
output of several hundred high-quality leaflets	
input of a hard copy image into a computer	

[5]

- (b) All of the uses in **part (a)** involve the input or output of data.

- (i) Describe **two** methods of preventing accidental loss of data.

1

2

[2]

- (ii) Describe **one** way of ensuring the security of the data against malicious damage.

[1]

67. M/J 15/P12/Q4/b

- (b) Sensors are used to monitor seismic activity. At the end of each day, all the data are transmitted to a central computer. This is hundreds of kilometres away.

Describe **one** way of ensuring that the integrity of the data is retained during the transmission stage.

[4]

68. M/J 15/P12/Q6/b

- 6 (a) Name the **most** suitable input or output device for each of the following uses.

Give a different device in **each** case.

Description of use	Input or output device
input of credit card number into an online form	
selection of an option at an airport information kiosk	
output of a single high-quality photograph	
output of several hundred high-quality leaflets	
input of a hard copy image into a computer	

[5]

- (b) All of the uses in **part (a)** involve the input or output of data.

- (i) Describe **two** methods of preventing accidental loss of data.

1

2

[2]

- (ii) Describe **one** way of ensuring the security of the data against malicious damage.

[1]

69. M/J 15/P13/Q3/a,b,c

- 3 (a)** Give the definition of the terms firewall and authentication. Explain how they can help with the security of data.

Firewall

Authentication

[3]

- (b)** Describe **two** differences between data integrity and data security.

[2]

- (c)** Data integrity is required at the input stage and also during transfer of the data.

- (i)** State **two** ways of maintaining data integrity at the input stage. Use examples to help explain your answer.

[3]

- (ii)** State **two** ways of maintaining data integrity during data transmission. Use examples to help explain your answer.

[3]

70. M/J 15/P23/Q4/b(ii)

- 4** A programming language has the built-in function `CONCAT` defined as follows:

```
CONCAT(String1 : STRING, String2 : STRING [, String3 : STRING] ) RETURNS STRING
For example: CONCAT("San", "Francisco") returns "SanFrancisco"
              CONCAT("New", "York", "City") returns "NewYorkCity"
```

The use of the square brackets indicates that the parameter is optional.

- (b)** A country has a number of banks. There are cash dispensers all over the country. Each bank is responsible for a number of dispensers.

- banks have a three digit code in the range 001 – 999
- each dispenser has a five digit code in the range 00001 – 99999

A text file, `DISPENSERS`, is to be created.

It has one line of text for each dispenser. For example: 00342▼007.

This line in the file is the data for dispenser 00342 which belongs to bank 007.

Incomplete pseudocode follows for the creation of the file `DISPENSERS`.

For the creation of the file, data is entered by the user at the keyboard.

- (ii)** No attempt has been made to validate the data entered by the user.

Describe **two** different types of validation check for the data entry.

1

2

[2]

Answers Section

1. M/J 26/P11/Q4/d

4(d)(i)	1 mark for each correct point to max 4 - Validation only ensures that data meets certain criteria - Validation prevents the input of inappropriate data - Verification makes sure that the data that is entered matches the original data / the data that was intended to be entered - Verification prevents data being accidentally changed from the original - Any example of validation/verification in practice e.g. entering a password twice	4	Accept a type of validation for MP1 BOD or a type of verification for MP3 BOD but not for both
4(d)(ii)	1 mark for a correct answer The original data was incorrect // the same wrong data was entered twice, and the wrong data was also valid // by example	1	User error is TV

2. M/J 26/P12/Q2/a,b,c

2(a)	1 mark for method 1 mark for correct matching description (* 2) - Visual check - The entered data is compared with the original by a person - Double entry - Enter the data twice and the computer automatically compares if they match	4	Note: Question asks for methods of data verification not validation Accept examples of visual checks, such as printing a copy and comparing the printout to the original Data as intended is NE. It is the comparison to the original that is important
2(b)	1 mark for two correct names of methods 1 mark for each correct matching use to max 2 - Presence check - The Date of Birth (DoB) field cannot be left blank - Range check - DoB must be after one date and before another date // between specific dates – must include both ends - Limit check - DoB must be after or before a specific date – must be only one end - Format check - DoB must be dd/mm/yyyy - Length check - DoB must be exactly 10 characters - Type check - The data type must be DATE or DATE/TIME	3	Note: Question asks for methods of data validation not verification Mark the first answer in each box The use must match the example given in the question 05/07/2010 Do not accept range and limit as two different checks Take care that answers do not just repeat the method, must be applied in some way. The use must match the method If no methods are given read the uses which must be applied to the DoB. For format check accept mm/dd/yyyy
2(c)(i)	1 mark for correct description of a virus Description: malicious software that replicates on the computer and can fill storage or delete/corrupt/change data - cause harm is NE 1 mark each for a correct method and matching description, 2 marks Method: Anti-virus software // anti-malware software Description: The software scans files/downloads/device and compares against database of known viruses, and quarantines/deletes/alerts the user if a match found	3	The description of the method must match the method chosen

2(c)(ii)	1 mark for correct description of a hacker Description: person who gains unauthorised access to a computer system 1 mark each for a correct method and matching description, 2 marks For example: Method: Firewall Description: The firewall monitors all incoming and outgoing data compares it to stored criteria and blocks transmissions that are not allowed Method: Multi-factor authentication Description: A second form of verification such as biometrics or use of an authenticator app is required for logging in Method: <u>Strong</u> passwords Description: - The password contains a mixture of different types of characters and is at least a set number of characters and is not something connected to the individual such as a date of birth Method: Encryption Description: Scramble/encode the data/turn into ciphertext so that even if it is accessed it will not be understood/useable without the decryption key	3 The description of the method must match the method chosen Just malicious intent is NE for unauthorised access Accept two-factor for multi-factor If firewall is the candidate's answer for question 2(c)(i) ensure it is not repeated here Just authentication is NE for multi-factor authentication Map biometrics/use of biometric password as the method to strong password Description: Uses features of an individual that are unique and which are almost impossible to duplicate
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3. M/J 26/P13/Q1

1	1 mark for each correct term Visual check is a method of verification where the data is entered by a user and then manually compared to the original data to make sure that they are the same. A verification method used during data transfer calculates a value from the data which is transferred with the data. The receiver performs the same calculation and compares the result with the value transferred. This method of verification is checksum. There are several methods of data validation. Existence check compares the data entered to a list of possible values that can be entered. If the data is not one of these values, it is rejected. Format check can be used to make sure data entered has the correct combination of letters, numbers and symbols. For example, a code may need to be two capital letters, one number followed by one symbol. Length check is used to make sure the correct quantity of characters has been entered. For example, a name must be more than one character and less than or equal to 100 characters.	5
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4. O/N 25/P11/Q8/a

8(a)	1 mark for each correct explanation, max 2 marks Virus: Malicious program/software that replicates/copies itself and deletes/alters files/data stored on a computer Pharming: Malicious code/software installed on a computer which redirects user to a fake website to obtain personal data	2
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5. O/N 25/P12/Q1

1	1 mark for 2 or 3 correctly connected verification method boxes, 2 marks for all 4 correct Verification Method Use <pre>graph LR; subgraph Verification_Method [Verification Method]; P1[Parity byte check]; C[Checksum]; V[Visual check]; P2[Parity block check]; end; subgraph Use [Use]; DT[Data transfer]; DE[Data entry]; end; P1 --- DT; C --- DT; V --- DE; P2 --- DT;</pre> The diagram consists of two columns. The left column, titled 'Verification Method', contains four boxes: 'Parity byte check', 'Checksum', 'Visual check', and 'Parity block check'. The right column, titled 'Use', contains two boxes: 'Data transfer' and 'Data entry'. Lines connect the boxes as follows: 'Parity byte check' to 'Data transfer'; 'Checksum' to 'Data transfer'; 'Visual check' to 'Data entry'; and 'Parity block check' to 'Data transfer'.	2
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6. O/N 25/P12/Q5/e

5(e)	1 mark for correct threat 1 mark corresponding description 1 mark for correct method of prevention <table border="1"> <thead> <tr> <th>Threat</th><th>Description</th><th>Prevention</th></tr> </thead> <tbody> <tr> <td>Malware / Virus</td><td>Malicious code that can alter/delete files</td><td>Anti-virus // Anti-malware // Firewall</td></tr> <tr> <td>Spyware</td><td>Records keystrokes which are sent to third party</td><td>Anti-spyware // Firewall // Anti-malware</td></tr> <tr> <td>Hacking</td><td>Gaining <u>unauthorised</u> access to a computer network / device</td><td>Authentication // Firewall</td></tr> <tr> <td>Phishing</td><td><u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information</td><td>Use a spam filter // Do not open emails from unknown sources</td></tr> <tr> <td>Pharming</td><td>Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information</td><td>Use a VPN // Anti-malware// Do not open links or download attachments</td></tr> </tbody> </table>	Threat	Description	Prevention	Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall	Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware	Hacking	Gaining <u>unauthorised</u> access to a computer network / device	Authentication // Firewall	Phishing	<u>Emails</u> supposedly from reputable companies are sent to trick people into revealing personal information	Use a spam filter // Do not open emails from unknown sources	Pharming	Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments	3
Threat	Description	Prevention																		
Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall																		
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Pharming	Users are directed to a bogus <u>website</u> that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments																		

7. O/N 25/P12/Q6/c

6(c)(i)	1 mark for correct answer parity bit ↓ <table><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>	0	1	0	1	1	1	0	0	1																																					
0	1	0	1	1	1	0	0																																								
6(c)(ii)	1 mark for correct answer parity bit ↓ <table><tr><td></td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td></td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>parity byte→</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>		1	0	1	1	0	1	1	1		0	1	1	1	0	0	0	0		0	0	0	1	1	0	1	1		0	1	1	1	0	1	0	0	parity byte→	1	0	1	0	0	0	0	0	1 [1]
	1	0	1	1	0	1	1	1																																							
	0	1	1	1	0	0	0	0																																							
	0	0	0	1	1	0	1	1																																							
	0	1	1	1	0	1	0	0																																							
parity byte→	1	0	1	0	0	0	0	0																																							

8. O/N 25/P12/Q10/c

10(c)	1 mark for a correct answer • Digital Signature	1
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9. O/N 25/P13/Q7/c,e

7(c)	1 mark for the name 2 marks for the matching description, max 3 marks - Parity Byte Check - a parity bit is added to each byte to make the number of 1s match the parity, odd or even - each byte can be checked on receipt and request to be resent if the byte does not match parity - Parity Block Check - a bit is added to each byte, but a parity byte is also set for each block - the location of an error can be found using vertical and horizontal parity - Checksum - A calculation is made from the data and transmitted with the data - The receiver performs the same calculation and compares with received checksum to see if they match	3
7(e)	1 mark for each validation check correctly identified, max 3 marks Range Check Existence Check Format Check	3

10. O/N 25/P21/Q6/a

6(a)	Number: Any number where the first two digits sum to 5 / 15 and the last three digits are 623 Explanation: The remainder is the same / 3 // The sum of the (first four) digits is the same / 13	2
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11. M/J 25/P11/Q7/a,b,c

7(a)	1 mark for a correct method: 1 mark each to max 2 for a corresponding description Method: Encryption Description: - Data is encoded/scrambled using a key to create cipher text - if intercepted it cannot be understood - without being decrypted using a key	3
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7(b)	1 mark each to max 5 • The sender hashes the document / message • to produce a digest • The sender encrypts the digest to create the digital signature • The message and the signature are sent to the banker / receiver • The receiver decrypts the signature to reproduce the digest • The receiver uses the same hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document has not changed	5
7(c)	1 mark each to max 3 • The data is put through an algorithm to create a checksum value • The data and checksum are sent to the receiver • The receiver performs the same algorithm on the data • if both checksums match the data is verified	3

12. M/J 25/P13/Q3/b

3(b)	1 mark for method and 1 mark for corresponding description During transmission e.g. • Encryption // by example such as VPN • Jumble / encode data so it cannot be decrypted/understood without the key On computer e.g. • Firewall / proxy • Filter incoming transmissions and stop any that could be attempting unauthorised access • Anti-malware • Find and delete or quarantine any malware that could delete the data / files • Encryption • Jumble / encode data so it cannot be decrypted / understood without the key • Physical method // by example • For example, the computer storing the data cannot be accessed without the key to the room	4
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13. O/N 24/P12/Q4/c

4(c)	1 mark for the security method. 2 marks for explanation Security method: Encryption Explanation • File contents are converted to cipher text • If intercepted the data cannot be understood without the decryption key	3
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14. M/J 24/P11/Q5/b,c(i)

5(b)	1 mark for each correctly completed term: • odd or even • 7-bits • odd • block • byte Computer A and Computer B agree on whether to use odd or even parity. Computer A divides the data into groups of 7-bits. The number of 1s in each group is counted. If the agreed parity is odd and the group has an even number of 1s a parity bit of 1 is appended, otherwise a parity bit of 0 is appended. In a parity block check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity byte.	5
5(c)(i)	1 mark each to max 3: • Compares all incoming and outgoing transmissions • ... against set criteria/whitelist/blacklist • Blocks all transmissions that do not meet rules • Blocks data entering from specific ports • Blocks unauthorised/unknown internal software transmitting data	3

15. M/J 24/P12/Q3/a

3(a)(i)	1 mark for security measure 1 mark each to max 2 for how the chosen measure works: • Firewall • Checks incoming connections • ... against criteria • Blocks data from entering specific ports • Blocks data that does not meet whitelist that meets blacklist • Proxy server • Prevents devices accessing the web server directly • Intercepts any requests • Forwards the request using its own IP address • Screens returning data before sending it to the user	3
3(a)(ii)	1 mark for security measure 1 mark each to max 2 for description of the chosen measure: • Encryption • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	3

16. M/J 24/P13/Q7/f

7(f)(i)	1 mark for each correct method and 1 mark for corresponding description to max 4 : <table><tr><th>Method</th><th>Description</th></tr><tr><td>Parity byte</td><td>An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.</td></tr><tr><td>Parity block</td><td>Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.</td></tr><tr><td>Checksum</td><td>A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.</td></tr></table>	Method	Description	Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.	Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.	Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.	4
Method	Description									
Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.									
Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.									
Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.									
7(f)(ii)	1 mark each to max 2 : • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	2								

17. O/N 23/P12/Q5/a,b,c

5(a)	1 mark for: Either - Ensuring data can only be accessed by / disclosed to authorised persons Or - Ensuring data cannot be accessed by / disclosed to unauthorised persons	1
5(b)	1 mark for each bullet point (max 1) - Ensuring the accuracy / completeness / consistency of data (during / after processing) - Ensuring the data is up to date	1
5(c)	1 mark for each bullet point. Phishing email (max 2) - The email pretends to be from an official body ... persuading individuals to disclose private information // by example such as bank details ... or requesting authentication by redirecting to an unofficial/unauthorised website // inviting a user to click a link Spyware (max 2) - Malware downloaded without the user's knowledge ... which secretly records the user's actions / keystrokes on the computer ... and sends logs of the actions to a third party	4

18. O/N 23/P13/Q3/a(iii),c

3(a)(iii)	1 mark each for firewall, encryption and passwords. Firewall: - Monitors incoming and outgoing traffic and rejects any traffic that does not meet the set rules Encryption: - Ensures that if data is intercepted / obtained it cannot be understood without the decryption key Passwords: - Ensures only users with the correct password can access the resources // prevents unauthorised access	3
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3(c)	1 mark for each bullet point. - The administrator completes a visual check / checks by eye ...that the tutor identifier input matches the tutor identifier on the original document - Double entry check // the administrator (or a second person) enters the number a second time ...and the system compares it with the first entry	4
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19. O/N 23/P13/Q8/a,b,c

8(a)	1 mark for each bullet point (max 2) - Validation // a validation method named or described ...protects the data by ensuring that the data is reasonable / sensible and within specified bounds	2
8(b)	1 mark for difference 1 mark for similarity Difference: - Pharming is malicious code that redirects to a fake website. Phishing uses an email to prompt user action. - Pharming is automatic. Phishing requires user action. Similarity: - Both try to obtain financial or personal information - Both are a false representation of an official organisation, e.g. a bank - Both make use of fake websites	2
8(c)	1 mark for each bullet point (max 3). - Download programs from reputable websites / sources ...as these are less likely to contain malware - Backup / archive computer systems ...so they can be restored in case of data loss from malware program installation Install and run anti-malware program ...so that regular scans can be made for known malware ...and if malware is found it can be quarantined / removed ...and computer's anti-malware definitions are regularly updated - Using a firewall to block unused ports ...so that malware cannot enter the computer system - Deny administrator privileges to everyday users ...so that malware cannot be downloaded by everyday users - Avoid the use of / access to removable devices ...so that malware cannot be installed from these devices	3

20. M/J 23/P12/Q2/c(i)

2(c)(i)	1 mark each to max 2 • Presence check to make sure that the (rider level) is entered • Look-up / Existence check to make sure the rider level is only Beginner, Intermediate or Advanced • Length check to make sure the rider level entered is either 8 or 12 characters • Type check to make sure the rider level is alphanumeric	2
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21. M/J 23/P13/Q4/d

4(d)(i)	1 mark each to max 2 • Length check: the registration number must be 6 characters long • Format check: the registration number must be in the format letter-digit-digit-digit-letter-letter • Type check: the registration number must be alphanumeric	2
4(d)(ii)	1 mark each • Visual check: Manually compare the registration number entered with the source document • Double entry: Enter the registration number twice and the computer compares to check they are the same	2
4(d)(iii)	The registration number on the original document might be in the correct format but may be the incorrect registration number for that car.	1

22. M/J 23/P13/Q6/a,b

6(a)	1 mark each to max 5 • The sender hashes the document • ... to produce a <u>digest</u> • The sender <u>encrypts</u> the digest to create the digital signature • The message and the signature are sent to the receiver • The receiver <u>decrypts</u> the signature to reproduce the digest • The receiver uses the <u>same</u> hashing algorithm on the document received to produce a second digest • The receiver compares this digest with the one from the digital signature • If both of the receiver's digests are the same the document is authentic	5
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6(b)	1 mark each for identification and appropriate description of 2 pieces of software, max 2 <table><tr><th>Type of software</th><th>Description</th></tr><tr><td>Antivirus</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr><tr><td>Antispyware</td><td> - scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing. </td></tr><tr><td>Firewall</td><td> - monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria </td></tr><tr><td>Antimalware</td><td> - scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing </td></tr></table>	Type of software	Description	Antivirus	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	Antispyware	- scans the computer for spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known spyware and prevents the download continuing.	Firewall	- monitors incoming and outgoing traffic and compares it to criteria that are set by the user such as through a whitelist/blacklist/identifying allowed / blocked IP addresses - compares incoming and outgoing traffic to criteria blocks those that do not match criteria	Antimalware	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing	2
Type of software	Description											
Antivirus	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing											
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Antimalware	- scans the computer for viruses and checks against a stored database of viruses, that needs to be updated regularly and then deletes / quarantines them - compares downloaded files to a database of known viruses and prevents the download continuing											

23. M/J 23/P23/Q2/b

2(b)(i)	MP1 Value for <u>month</u> is between 1 and 12 (inclusive) MP2 Value of <u>year</u> is <= 2002	2
2(b)(ii)	MP1 Reference to <u>month</u> and <u>day</u> MP2 Clear description for a <u>check</u> that the day number matches with a relevant month (Either day matches with month // month matches with day)	2

24. O/N 22/P11/Q2

2	1 mark for each correct line. Security feature firewall pharming anti-virus software encryption converts data to an alternative form redirects a user to a false website verifies the authenticity of data scans files on the hard drive for malicious software accepts or rejects incoming and outgoing packets based on criteria	4
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25. O/N 22/P12/Q4/a,b,c

4(a)	data verification is checking if input data is the same as the original whereas data validation is checking that the data is reasonable / sensible	1
4(b)	1 mark for each bullet point (max 3): - checksum value is calculated from the data before transmission // correct description of a checksum algorithm ... this calculated value is transmitted with the data receiving computer recalculates the checksum from the received data - if the checksum received and calculated match, no error has occurred // if the checksum received and calculated do not match, an error has occurred	3
4(c)	1 mark for each bullet point (max 2). For example: - to make sure data is in the required format // only expected characters allowed - to make sure the data is already present in the system - to make sure the data contains the correct number of characters - to ensure that non-numeric data is entered	2

26. O/N 22/P12/Q6/a

6(a)(i)	1 mark for each method of ensuring authenticity (max 2): (email) message put through hashing algorithm to produce a digest - Digest encrypted with <u>sender's private</u> key (to create the digital signature) - the (digital) signature can only be decrypted with matching <u>sender's public</u> key	2
6(a)(ii)	1 mark for each bullet point: monitors <u>incoming</u> and <u>outgoing</u> packets / traffic checks against an allow list / deny list of IP addresses // checks against a set of rules for acceptable data / ports etc. blocks transmissions that do not meet criteria / rules // allows through is satisfies the criteria /rules	3

27. O/N 21/P11/Q2/a,b,c

2(a)	1 mark per bullet point - security is protecting data from loss / corruption - integrity is ensuring the consistency / accuracy of the data	2
2(b)(i)	1 mark per bullet point - validation checks that data is reasonable / sensible - example e.g. checking data is the right number / type of characters	2
2(b)(ii)	1 mark per bullet point - verification checks that data is the same as the original - by example e.g. double entry	2
2(c)	1 mark per similarity to max 2 - Both are pieces of malicious software - Both are downloaded / installed/run without the user's knowledge - Both can pretend to be / are embedded in other legitimate software when downloaded // both try to avoid the firewall - Both run in the background 1 mark for difference - Virus can damage computer data; spyware only records / accesses data - Virus does not send data out of the computer; spyware sends recorded data to third party - Virus replicates itself; spyware does not replicate itself	3

28. O/N 21/P12/Q1

1	1 mark for 3 correct lines only from Data Security 1 mark for 2 correct lines only from Data Integrity <pre> graph LR Firewall --> DS[Data Security] DoubleEntry[Double entry] --> DS PresenceCheck[Presence check] --> DI[Data Integrity] AccessRights[Access rights] --> DI Password --> DS </pre>	2
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29. O/N 21/P13/Q2/a,b,c

2(a)	1 mark per bullet point • security is protecting data from loss / corruption • integrity is ensuring the consistency / accuracy of the data	2
2(b)(i)	1 mark per bullet point • validation checks that data is reasonable / sensible • example e.g. checking data is the right number / type of characters	2
2(b)(ii)	1 mark per bullet point • verification checks that data is the same as the original • by example e.g. double entry	2
2(c)	1 mark per similarity to max 2 • Both are pieces of malicious software • Both are downloaded / installed/run without the user's knowledge • Both can pretend to be / are embedded in other legitimate software when downloaded // both try to avoid the firewall • Both run in the background 1 mark for difference • Virus can damage computer data; spyware only records / accesses data • Virus does not send data out of the computer; spyware sends recorded data to third party • Virus replicates itself; spyware does not replicate itself	3

30. M/J 21/P11/Q6/a,b,c

6(a)	Range (check)	1
6(b)	Presence (check)	1
6(c)	Existence (check)	1

31. M/J 21/P12/Q8/a,b,c

8(a)	1 mark per bullet point • Security protects data against loss • Privacy protects data against unauthorised access	2
8(b)	1 mark for a correct answer • Two factor authentication • Biometric passwords • Key Card Access • Firewall	1
8(c)	1 mark per correct answer to max 2 • Malware // viruses // spyware // by example • Hacking • Phishing • Pharming	2

32. M/J 21/P13/Q6/a,b,c

6(a)	Range (check)	1
6(b)	Presence (check)	1
6(c)	Existence (check)	1

33. 9608 M/J 21/P11/Q2/d

2(d)	1 mark per bullet point to max 3 • Password • Biometrics • Access rights • Swipe cards • Physical access measures, e.g. security guards • Implement a firewall/proxy to monitor remote access requests • Two-step authentication	3
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34. 9608 M/J 21/P12/Q5/e(ii)

5(e)(ii)	1 mark for each (max 2) • range check • length check • type check	2
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35. 9608 M/J 21/P12/Q8/a,b

8(a)	1 mark per term <table><tr><th>Description</th><th>Term</th></tr><tr><td>Ensures the data is accurate and up to date</td><td>Data integrity</td></tr><tr><td>Prevents accidental or malicious data loss</td><td>Data security</td></tr><tr><td>Prevents unauthorised access to data</td><td>Data privacy</td></tr></table>	Description	Term	Ensures the data is accurate and up to date	Data integrity	Prevents accidental or malicious data loss	Data security	Prevents unauthorised access to data	Data privacy	3
Description	Term									
Ensures the data is accurate and up to date	Data integrity									
Prevents accidental or malicious data loss	Data security									
Prevents unauthorised access to data	Data privacy									
8(b)	1 mark per bullet point to max 2 • Mathematical algorithm // encrypted data • Attached to electronically transmitted document • ...to verify its content/that it comes from a trusted source/	2								

36. 9608 M/J 21/P13/Q1/a,b

1(a)	1 mark for each validation type <table border="1"> <thead> <tr> <th>Field</th><th>Validation description</th><th>Validation type</th></tr> </thead> <tbody> <tr> <td>Name</td><td>A name must be entered</td><td>Presence check</td></tr> <tr> <td>Date of birth</td><td>Entered as dd/mm/yyyy</td><td>Format check</td></tr> <tr> <td>Telephone number</td><td>A limit of 15 characters can be entered</td><td>Length check</td></tr> <tr> <td>Experience level</td><td>Only values between 1 and 5 can be entered</td><td>Range check</td></tr> </tbody> </table>	Field	Validation description	Validation type	Name	A name must be entered	Presence check	Date of birth	Entered as dd/mm/yyyy	Format check	Telephone number	A limit of 15 characters can be entered	Length check	Experience level	Only values between 1 and 5 can be entered	Range check	4
Field	Validation description	Validation type															
Name	A name must be entered	Presence check															
Date of birth	Entered as dd/mm/yyyy	Format check															
Telephone number	A limit of 15 characters can be entered	Length check															
Experience level	Only values between 1 and 5 can be entered	Range check															
1(b)(i)	Verification // by example	1															
1(b)(ii)	1 mark per bullet point: • Parity • Checksum	2															

37. 9608 M/J 21/P21/Q5/a(ii)

5(a)(ii)	One mark per point. Check for: - Integer / number out of range <1 OR > 20 - Real number entered - Non-numeric value entered Max 2 marks	2
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38. M/J 20/P11/Q1/a,b

1(a)	1 mark for each correctly completed term. Validation checks that the data entered is reasonable. One example is a presence check. Verification checks that the data entered is the same as the original. One example is double entry.	4								
1(b)	1 mark for each correct entry <table><tr><th>Security measure</th><th>Description</th></tr><tr><td>Disk mirroring</td><td>Data are written on two or more disks simultaneously.</td></tr><tr><td>Encryption</td><td>Contents are scrambled so they cannot be understood without a decryption key</td></tr><tr><td>Backup</td><td>A copy of the data is taken and stored in another location</td></tr></table>	Security measure	Description	Disk mirroring	Data are written on two or more disks simultaneously.	Encryption	Contents are scrambled so they cannot be understood without a decryption key	Backup	A copy of the data is taken and stored in another location	3
Security measure	Description									
Disk mirroring	Data are written on two or more disks simultaneously.									
Encryption	Contents are scrambled so they cannot be understood without a decryption key									
Backup	A copy of the data is taken and stored in another location									

39. M/J 20/P12/Q7/d

7(d)(i)	1 mark per bullet point to max 2 - Prevents unauthorised access to the data - Monitors incoming and outgoing traffic - Blocks transmissions from unauthorised sources / websites / ports - Maintains an allow list / deny list of IP addresses	2
7(d)(ii)	1 mark only from: - Running up-to-date antivirus software - Use of proxy server - Strong / Biometric passwords - Etc.	1

40. M/J 20/P13/Q5/a

5(a)	1 mark per bullet point to max 3 • Encryption scrambles the source code (so it is meaningless) • ... using an encryption key / algorithm • If the file is accessed without authorisation it will be meaningless • It requires a decryption key / algorithm to unscramble	3
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41. O/N 19/P13/Q1/c(ii)

1(c)(ii)	1 mark for each security method to max 2 , 1 mark for integrity Security • encryption • access rights • username and password // biometrics // user accounts • backup // disk mirroring • firewall • Physical methods (e.g. CCTV, locked rooms etc.) Integrity • checksum • parity • validation on input	3
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42. M/J 19/P12/Q5/b(i)

5(b)(i)	1 mark per bullet point • Security ensures that data is safe from unauthorised access // safe from loss • Integrity ensures that data is accurate / consistent / up to date	2
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43. M/J 19/P13/Q2/a,b,c

2(a)	1 mark per bullet point • Security: keep data safe from accidental/malicious damage/loss • Privacy: keep data confidential // only seen by authorised personnel	2
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2(b)	1 mark for identifying method 2 marks for description to max 2 × 3 For example: • User accounts • User has a username and password Access to resources can be limited to specific accounts • Cannot access <u>system</u> without valid username and password // prevents unauthorised access to the <u>system</u> • Firewall • All incoming and outgoing network traffic goes through firewall • Blocks signals that do not meet requirements • Keeps a log of signals • Applications can have network access restricted • Anti-malware • Scans for malicious software • Quarantines or deletes any malicious software found • Scans can be scheduled at regular intervals • Should be kept up to date • Auditing • Logging all actions/changes to the system • In order to identify any unauthorised use • Application Security (accept equivalent) • Applying regular updates / patches • Finding, fixing and preventing security vulnerabilities in any (installed) application	6
2(c)(i)	1 mark per bullet to max 4 • Each byte has a parity bit // horizontal parity • An additional parity byte is sent with vertical (and horizontal) parity • Each row and column must have an <u>even/odd number of 1s</u> • Identify the incorrect row and column • The intersection is the error	4
2(c)(ii)	1 mark for correct answer • Errors in an even number of bits (could cancel each other out)	1

44. O/N 18/P12/Q6/a

6(a)	1 mark for each method to max 2 • Biometric authentication // by example • Two-step authentication // by example • Firewall / proxy • Encryption • Different access rights for different users • Password protect <u>file</u> (using a different password) • Anti-malware	2
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45. O/N 18/P13/Q5/b

5(b)	1 mark per bullet point, max 3 • Firewall / proxy • Encryption • Username and Password • Physical security • Biometric authentication // by example • Two-step authentication // by example • Anti-malware	3
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46. M/J 18/P11/Q6/a,b,c,d

6(a)	1 mark for the correct parity Parity bit <table><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>	0	0	1	0	0	0	0	0	1																																									
0	0	1	0	0	0	0	0																																												
6(b)(i)	1 mark for the correct bit circled <table><tr><td>Parity bit</td><td colspan="7">Data</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Parity byte</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr></table>	Parity bit	Data							1	1	0	1	0	1	1	1	1	0	0	0	1	1	1	0	0	1	0	0	1	0	1	1	1	1	1	0	1	1	1	1	Parity byte	1	1	1	1	1	0	0	1	1
Parity bit	Data																																																		
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1	0	0	0	1	1	1	0																																												
0	1	0	0	1	0	1	1																																												
1	1	1	0	1	1	1	1																																												
Parity byte	1	1	1	1	1	0	0	1																																											
6(b)(ii)	1 mark for each bullet point • row and column has incorrect parity (odd instead of even) • the intersection identifies the error	2																																																	
6(c)	1 mark per bullet to max 2 • Errors in an even number of bits (in the same row or column) • Prevents error being identified • Could appear to be correct	2																																																	

6(d)	1 mark for the name and max 2 marks for description • Check sum • A calculation is done on a block of data • The result is transmitted with the data • Calculation repeated at receiving end • Results compared • If different an error has occurred • Hash total • Total of several fields of data • Including fields not usually used in calculations • The result is transmitted with the data • Calculation repeated at receiving end • Results compared • If different an error has occurred	3
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47. M/J 18/P12/Q3/a,b,c

3(a)	1 mark for correct parity bit Parity bit <table><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>	0	0	1	0	0	0	0	0	1																																														
0	0	1	0	0	0	0	0																																																	
3(b)	1 mark for the correct bit circled. <table><tr><td></td><td>Parity bit</td><td colspan="7">Data</td></tr><tr><td></td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td></td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td></td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td></td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Parity byte</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr></table>		Parity bit	Data								1	0	1	0	1	1	1	1		0	1	1	0	0	1	1	0		1	1	0	0	0	0	0	0		0	1	0	0	0	0	0	0	Parity byte	0	0	0	0	1	0	0	1	1
	Parity bit	Data																																																						
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	0	1	1	0	0	1	1	0																																																
	1	1	0	0	0	0	0	0																																																
	0	1	0	0	0	0	0	0																																																
Parity byte	0	0	0	0	1	0	0	1																																																
3(c)	1 mark per each correct row <table><tr><td>Measure</td><td>Validation</td><td>Verification</td></tr><tr><td>Checksum</td><td></td><td>✓</td></tr><tr><td>Format check</td><td>✓</td><td></td></tr><tr><td>Range check</td><td>✓</td><td></td></tr><tr><td>Double entry</td><td></td><td>✓</td></tr><tr><td>Check digit</td><td>✓</td><td></td></tr></table>	Measure	Validation	Verification	Checksum		✓	Format check	✓		Range check	✓		Double entry		✓	Check digit	✓		5																																				
Measure	Validation	Verification																																																						
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Format check	✓																																																							
Range check	✓																																																							
Double entry		✓																																																						
Check digit	✓																																																							

48. M/J 18/P13/Q4/a,b,c

4(a)	Parity bit <table><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>	1	0	1	1	1	0	1	0	1																																														
1	0	1	1	1	0	1	0																																																	
4(b)	1 mark for correctly circled bit <table><tr><td></td><td>Parity bit</td><td colspan="7">Data</td></tr><tr><td></td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td></td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td></td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td></td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>Parity byte</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td></tr></table>		Parity bit	Data								0	0	1	1	0	1	0	1		1	1	1	1	1	0	0	1		1	0	1	0	1	0	0	0		0	0	0	1	1	0	1	1	Parity byte	0	1	1	1	1	1	0	1	1
	Parity bit	Data																																																						
	0	0	1	1	0	1	0	1																																																
	1	1	1	1	1	0	0	1																																																
	1	0	1	0	1	0	0	0																																																
	0	0	0	1	1	0	1	1																																																
Parity byte	0	1	1	1	1	1	0	1																																																
4(c)	1 mark for each correct line Error detection measure Type check Proof reading Check digit Checksum Verification Validation	4																																																						

49. O/N 17/P12/Q3/a,b,c

3(a)(i)	1 Mark per bullet, max 3 • Security is keeping the data safe • From accidental / malicious damage /loss • By example of need for security • Privacy is the need to restrict access to personal data • To avoid it being seen by unauthorised people • By example of need for privacy	3
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3(a)(ii)	1 Mark for a suitable example For example: Personal data of students / staff	1
3(b)	1 Mark for stating the security measure 1 Mark for a corresponding description Maximum 2 marks for each measure Maximum 2 measures Physical measures • Locked doors/keyboards etc. • Secure methods of access, keypads/ biometric scans etc. Backup of data • Regular copies of the data are made • If the data is corrupted it can be restored Disk-mirroring • All activity is duplicated to a second disk in real time so that if the first disk fails there is a complete copy available Access rights • Different access rights for individuals/groups of users • To stop users editing data they are not permitted to access • By example Encryption • If accessed, data cannot be understood by unauthorised personnel • Accessed only by those with the decryption key Firewall • To stop unauthorised access/hackers gaining access to the computer network Use authentication methods such as passwords and usernames • Passwords should be strong / biometrics • To prevent unauthorised access to data Anti-malware program • To detect / remove / quarantine viruses / key-loggers etc. • Carrying out regular scans Concurrent Access Controls // Record locking • Closes a record to second user until first update complete • To prevent simultaneous updates being lost	4
3(c)	1 Mark per bullet, max 2 • Checking that the data entered matches / is consistent with that of the source. • Comparison of two versions of the data • Examples include double entry, visual checking, proof reading etc... • In the event of a mismatch – the user is forced to re-enter the data • By example, e.g. creation of a password • Does not check data is sensible/acceptable	2

50. O/N 17/P13/Q3/a,b

3(a)	Two marks from: - Physical measures - Access rights - Encryption - Firewall - Use authentication methods such as usernames and passwords - Anti-malware program	Max 2																																													
3(b)(i)	<table><tr><td>7</td><td>X</td><td>6</td><td>=</td><td>42</td></tr><tr><td>8</td><td>X</td><td>5</td><td>=</td><td>40</td></tr><tr><td>6</td><td>X</td><td>4</td><td>=</td><td>24</td></tr><tr><td>5</td><td>X</td><td>3</td><td>=</td><td>15</td></tr><tr><td>3</td><td>X</td><td>2</td><td>=</td><td>6</td></tr><tr><td>1</td><td>X</td><td>1</td><td>=</td><td>1</td></tr><tr><td></td><td></td><td></td><td>Total:</td><td>128 / 11</td></tr><tr><td></td><td></td><td></td><td></td><td>11 R 7</td></tr><tr><td colspan="4">Check digit:</td><td>11 – 7 = 4</td></tr></table> 1 mark for 6 values 1 mark for 2 steps Accept 128 MOD 11 = 7 1 mark for subtraction Answer: 786531 4 (1 mark for answer)	7	X	6	=	42	8	X	5	=	40	6	X	4	=	24	5	X	3	=	15	3	X	2	=	6	1	X	1	=	1				Total:	128 / 11					11 R 7	Check digit:				11 – 7 = 4	4
7	X	6	=	42																																											
8	X	5	=	40																																											
6	X	4	=	24																																											
5	X	3	=	15																																											
3	X	2	=	6																																											
1	X	1	=	1																																											
			Total:	128 / 11																																											
				11 R 7																																											
Check digit:				11 – 7 = 4																																											
3(b)(ii)	One mark for name of check One mark for description Max two checks Uniqueness check Each PatientID must be unique Length check Each PatientID is exactly 7 characters Format check / Type check All 7 characters must be <u>digits</u> Presence check PatientID must be entered	Max 4																																													

51. M/J 17/P11/Q5/a,b

5(a)(i)	- Count the number of one bits in the <u>first seven</u> bit positions - Add a 0 or 1 to bit position 0, to make the count of one bits an <u>odd</u> number	1 1	2
5(a)(ii)	A = 1 B = 1		1

5(a)(iii)	Two from: - A parity bit is worked out for each <u>column</u> 1 - The computer checks the parity of each bit position in parity byte // the computer generates copy of the parity byte and <u>compares</u> 1 - If incorrect parity then there is an error in the data received // No parity error means no error in the data received 1 - The position of the incorrect bit can be determined 1	2																																																																																
5(b)(i)	<table><tr><th colspan="8">Bit position</th></tr><tr><th>7</th><th>6</th><th>5</th><th>4</th><th>3</th><th>2</th><th>1</th><th>0</th></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr></table>	Bit position								7	6	5	4	3	2	1	0	1	0	0	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	1	1	0	1	0	1	1	1	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1	0	0	0	2
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5(b)(ii)	Three from: - Consider each row in sequence 1 - Identify any row with incorrect parity 1 - Repeat the process for each column in sequence 1 - Identify where a row and column with incorrect parity intersect 1	3																																																																																

52. M/J 17/P11/Q7/c(v)

7(c)(v)	Two checks from: One mark for check and one mark for description - Range check 1 Check the number entered is (say) between 1 and 100 1 - Format check 1 Checks the product code is a particular format // Checks the number has digit characters only // by example 1 - Length check 1 The number of items has exactly five characters 1 - Existence check 1 To ensure the product code has been assigned 1	Max 4
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53. M/J 17/P12/Q6/c(v)

6(c)(v)	Two checks from: One mark for check and one mark for description - Format check RunnerID is three letter characters followed by two digit characters //Position is digit characters only - Length check RunnerID has exactly five characters - Range check The value for Position is between 1 and (say) 50 - Presence check The text box for RunnerID or Position is not empty - Existence check To ensure that RunnerID has been registered - Uniqueness check To ensure no two runners have the same number	Max 4 1 1 1 1 1 1 1 1
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54. M/J 17/P13/Q5/a,b

5(a)(i)	- Count the number of one bits in the <u>first seven</u> bit positions - Add a 0 or 1 to bit position 0, to make the count of one bits an <u>odd</u> number	1 1	2
5(a)(ii)	A = 1 B = 1		1
5(a)(iii)	Two from: - A parity bit is worked out for each <u>column</u> - The computer checks the parity of each bit position in parity byte // the computer generates copy of the parity byte and <u>compares</u> - If incorrect parity then there is an error in the data received // No parity error means no error in the data received - The position of the incorrect bit can be determined	1 1 1 1	2

5(b)(i)	<table><tr><th colspan="8">Bit position</th></tr><tr><th>7</th><th>6</th><th>5</th><th>4</th><th>3</th><th>2</th><th>1</th><th>0</th></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table> <table><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr></table>	Bit position								7	6	5	4	3	2	1	0	1	0	0	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	1	1	0	1	0	1	1	1	1	1	0	0	0	1	1	1	0	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1	0	0	0	2
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0	0	0	0	0	0	0	1																																																																											
0	1	0	1	1	0	0	0																																																																											
5(b)(ii)	Three from: - Consider each row in sequence - Identify any row with incorrect parity - Repeat the process for each column in sequence - Identify where a row and column with incorrect parity intersect	3																																																																																

55. M/J 17/P13/Q7/c(v)

7(c)(v)	Two checks from: One mark for check and one mark for description - Range check 1 Check the number entered is (say) between 1 and 100 1 - Format check 1 Checks the product code is a particular format // Checks the number has digit characters only // by example 1 - Length check 1 The number of items has exactly five characters 1 - Existence check 1 To ensure the product code has been assigned 1	Max 4
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56. O/N 16/P11/Q7/c

- (c) (i) • Hackers can still access the data (and corrupt it, change it or delete it)
 • Encryption simply makes data incomprehensible (without decryption key / algorithm) [2]
- (ii) Any **two** from:
- This is an explanation of data verification (not validation)
 - Data validation ensures that data is reasonable / sensible / within a given criteria
 - Original data may have been entered correctly but is not reasonable (e.g. age of 210) [2]

- (iii) • A password does not prevent unauthorised access, it makes it more difficult
- Password can be guessed (if weak) // Password can be stolen // A relevant example of misappropriation of password [2]

57. O/N 16/P13/Q7/c

- (c) (i) • Hackers can still access the data (and corrupt it, change it or delete it)
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(ii) Any **two** from:

- This is an explanation of data verification (not validation)
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- (iii) • A password does not prevent unauthorised access, it makes it more difficult
- Password can be guessed (if weak) // Password can be stolen // A relevant example of misappropriation of password [2]

58. M/J 16/P13/Q7/a,b

7 (a) **Four** from: [4]

- Security is keeping the data safe.
- Integrity is making sure that the data is correct / valid.
- Security is the prevention of data loss.
- Integrity ensures that the data received is the same as the data sent / data copied is the same as the original.
- Example of ensuring security, e.g. usernames and passwords, firewalls etc...
- Example of ensuring integrity, e.g. parity checks, double entry etc...

(b) Three pairs from:

[6]

- Installing a firewall and ensuring it is switched on.
- To stop unauthorised access / hackers gaining access to the bank's computer network.
- Use authentication methods such as passwords and usernames.
- Passwords should be strong / biometrics.
- Encrypt the data.
- So that if data is accessed it will be meaningless / only accessed by those with decryption key.
- Set up access rights...
- To stop users reading/editing data they are not permitted to access.
- Installing and running an up to date anti-malware program (anti-virus/anti-spyware etc.).
- To detect / remove / quarantine viruses / key-loggers etc.
- Make regular backups of the data.
- To separate device or off site to enable recovery if necessary.
- Employ measures for physical security.
- Example of a measure for physical security.

59. O/N 15/P11/Q9/a,b

9 (a) One mark for validation, one mark for verification.

validation

- check whether data is reasonable / meets given criteria

verification

- method to ensure data which is copied / transferred is the same as the original
- entering data twice and computer checks both sets of data
- check entered data against original document / source

[2]

(b) any **four** from:

- parity can be even or odd
- parity check uses the number of 1s in a binary pattern
- if there is an even / odd number of 1s, then the parity is even / odd
- following transmission ...
- parity of each byte checked
- a parity bit is used to make sure binary pattern has correct parity
- example: 1 0 0 1 0 1 1 1 has parity bit set to 1 in MSB since system uses odd parity (original data: 0 0 1 0 1 1 1 which has four 1 bits)

[4]

60. O/N 15/P11/Q10/a,b

10 (a) any **two** from:

- malicious code / software / program
- that replicates / copies itself
- can cause loss of data / corruption of data on the computer
- can cause computer to “crash” / run slowly
- can fill up hard disk with data

[2]

(b) any **two** from:

- checks for boot sector viruses when machine is first turned on
- when an external storage device is connected
- checks a file / web page when it is accessed / downloaded

[2]

61. O/N 15/P13/Q9/a,b

9 (a) One mark for validation, one mark for verification.

Validation

- check whether data is reasonable / meets given criteria

Verification

- method to ensure data which is copied / transferred is the same as the original
- entering data twice and computer checks both sets of data
- check entered data against original document / source

[2]

(b) Any **four** from:

- parity can be even or odd
- parity check uses the number of 1s in a binary pattern
- if there is an even / odd number of 1s, then the parity is even / odd
- following transmission ...
- parity of each byte checked
- a parity bit is used to make sure binary pattern has correct parity
- example: 1 0 0 1 0 1 1 1 has parity bit set to 1 in MSB since system uses odd parity (original data: 0 0 1 0 1 1 1 which has four 1 bits)

[4]

62. O/N 15/P13/Q10/a,b

10 (a) any **two** from:

- malicious code / software / program
- that replicates / copies itself
- can cause loss of data / corruption of data on the computer
- can cause computer to “crash” / run slowly
- can fill up hard disk with data

[2]

(b) any **two** from:

- checks for boot sector viruses when machine is first turned on
- when an external storage device is connected
- checks a file / web page when it is accessed / downloaded

[2]

63. O/N 15/P21/Q7/c

(c) Used to provide an integrity/verification check

[1]

Used as a checksum

[1]

The total can be recalculated by the receiving software

[1]

If any of the characters have been incorrectly transmitted the recalculated total and transmitted total will not match

[1]

[MAX 2]

64. O/N 15/P23/Q7/c

- (c) Used to provide an integrity/verification check [1]
 - Used as a checksum [1]
 - The total can be recalculated by the receiving software [1]
 - If any of the characters have been incorrectly transmitted the recalculated total and transmitted total will not match [1]
- [MAX 2]

65. M/J 15/P11/Q4/b

(b) 1 mark for name + **3 marks** for description

parity check

- uses even or odd parity which is decided before data sent
- each byte has a parity bit
- parity bit is set to 0 or 1 to make parity for byte correct
- after transmission, parity of each byte re-checked
- if it is different, then an error is flagged
- any reference to use of parity blocks/parity byte to (identify position of incorrect bit)

checksum

- a calculation is carried out on the data to be sent (checksum)
- the result is sent, along with data to recipient
- checksum is re-calculated at receiving end
- if both sums are the same, no error has occurred
- if the sums are different, the data has been corrupted during transmission
- request is sent to re-send data

[4]

66. M/J 15/P11/Q6/b**(b) (i)** Any **two** from:

- frequent (or equivalent) backup EITHER to secondary media/to 3rd party server/cloud/removable devices/continuous backup OR stored remotely
- disk-mirroring strategy/RAID
- UPS (uninterruptable power supply)/backup generator

[2]**(ii)** Any **one** from:

- protection of data (or equivalent) with passwords/using password and username for logging on include e.g. fingerprint scanning
- encryption
- installation and use of up to date anti-malware/anti-virus
- give different access rights to different users
- use a firewall,
- physical methods/lock doors and use secure entry devices/CCTV

[1]

6 (a)

Description of use	Input or output device
input of credit card number into an online form	Keyboard/keypad/numberpad
selection of an option at an airport information kiosk	touch screen
output of a single high quality photograph	ink jet printer
output of several hundred high quality leaflets	laser printer
input of a hard copy image into a computer	scanner

[5]

67. M/J 15/P12/Q4/b

(b) 1 mark for name + **3 marks** for description

parity check

- uses even or odd parity which is decided before data sent
- each byte has a parity bit
- parity bit is set to 0 or 1 to make parity for byte correct
- after transmission, parity of each byte re-checked
- if it is different, then an error is flagged
- any reference to use of parity blocks/parity byte to (identify position of incorrect bit)

checksum

- a calculation is carried out on the data to be sent (checksum)
- the result is sent, along with data to recipient
- checksum is re-calculated at receiving end
- if both sums are the same, no error has occurred
- if the sums are different, the data has been corrupted during transmission
- request is sent to re-send data

[4]

68. M/J 15/P12/Q6/b

(b) (i) Any **two** from:

- frequent (or equivalent) backup EITHER to secondary media/to 3rd party server/cloud/removable devices/continuous backup OR stored remotely
- disk-mirroring strategy/RAID
- UPS (uninterruptable power supply)/backup generator

[2]

(ii) Any **one** from:

- protection of data (or equivalent) with passwords/using password and username for logging on include e.g. fingerprint scanning
- encryption
- installation and use of up to date anti-malware/anti-virus
- give different access rights to different users
- use a firewall,
- physical methods/lock doors and use secure entry devices/CCTV

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Description of use	Input or output device
input of credit card number into an online form	Keyboard/keypad/numberpad
selection of an option at an airport information kiosk	touch screen
output of a single high quality photograph	ink jet printer
output of several hundred high quality leaflets	laser printer
input of a hard copy image into a computer	scanner

[5]**69. M/J 15/P13/Q3/a,b,c**

- 3 (a)** maximum of **two marks** for firewall description + maximum of **two marks** for authentication description

Firewall

- sits between the computer or LAN and the Internet/WAN and permits or blocks traffic to/from the network
- can be software and/or hardware
- software firewall can make precise decisions about what to allow or block as it can detect illegal attempts by specific software to connect to Internet
- can help to block hacking or viruses reaching a computer

Authentication

- process of determining whether somebody/something is who/what they claim to be
- frequently done through log on passwords/biometrics
- because passwords can be stolen/cracked, digital certification is used
- helps to prevent unauthorised access to data

[3]

(b) one mark for security, one mark for integrity:

- integrity deals with validity of data/freedom from errors/data is reasonable
- security deals with protection of data
- security protects data from illegal access/loss
- integrity deals with making sure data is not corrupted after, for example, being transmitted

[2]

(c) (i) one mark for each way of maintaining data security + one mark for an example/enhancement

- validation (to ensure data is reasonable)
- examples include range checks, type checks, length checks, ...
- verification (checks if data input matches original/if transmitted data matches original)
- can use double data entry or visual check/other methods such as parity checks
- doesn't check whether or not data is reasonable

[3]

(ii) one mark for each way of maintaining data integrity + one mark for an example/enhancement

- parity checking
- one of the bits is reserved as parity bit
- e.g. 1 0 1 1 0 1 1 0 uses odd parity
- number of 1s must be odd
- parity is checked at receiver's end
- a change in parity indicates data corruption
- check sum
- adds up bytes in data being sent and sends check sum with the data
- calculation is re-done at receiver's end
- if not the same sum then the data has been corrupted during transmission

[3]

70. M/J 15/P23/Q4/b(ii)

- (ii)**
- Bank code/ Dispenser code is digit characters only
 - Bank code is exactly 3 digits // Dispenser code is exactly 5 digits
 - Range check on Bank code between 1 and 999
// range check on dispenser code between 1 and 99999

Note: If no reference made to either Bank code or Dispenser code MAX 1

[max 2]